

CASE STUDY | NO-CODE AI

Teaching a Machine to Read Body Pose

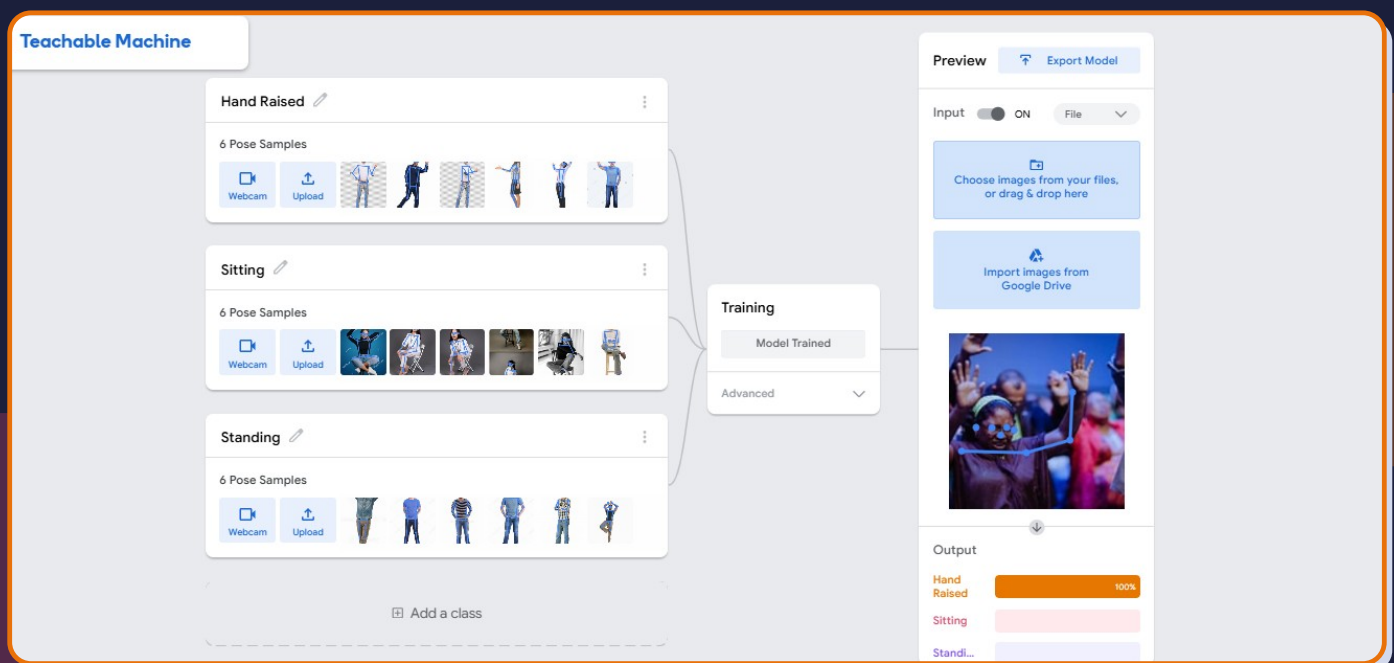
I trained an image model in Google's Teachable Machine to tell three poses apart (Hand Raised, Sitting, Standing) and tested it on photos it had never seen. This case study covers the build, the results and an honest look at what the test can and cannot prove.

Teachable Machine

Pose project

Image classification

No-code ML



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AI product thinking | Education and EdTech

Project Overview

What I built and how a model learns

Machine learning can sound like something only programmers do. Teachable Machine shows otherwise: you show the computer examples, press train, and it learns to recognise them. My goal was to build a small pose classifier and check it on new images.

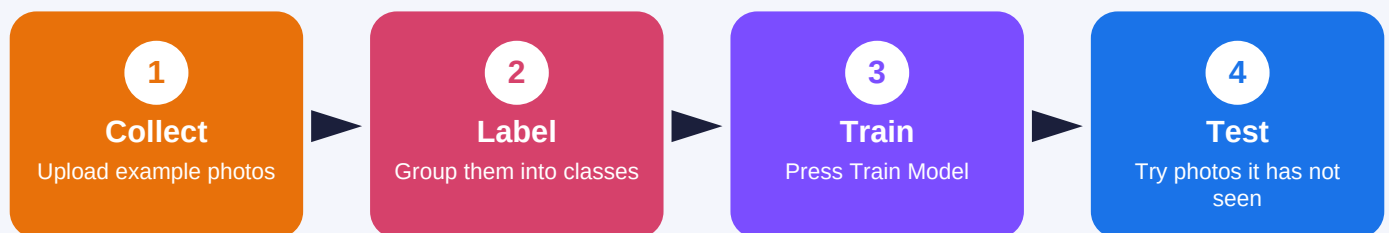
Objective

Train a pose model with at least 3 classes and 6 samples per class, then test it on images that were not in the training data.

Tool

Google Teachable Machine, Pose Project (teachablemachine.withgoogle.com). It finds body keypoints in a photo and learns from their positions.

How the model learns, in four steps



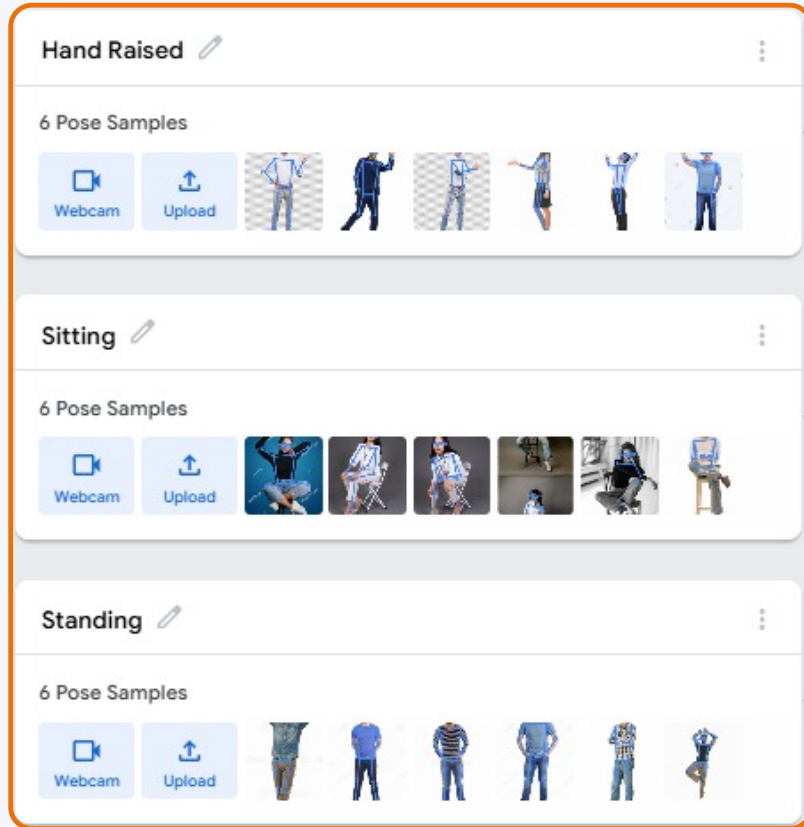
A teacher's way to see it

Training a model is like teaching a new student with flashcards. The model only knows what the flashcards show. If the examples are narrow, it will be confident but wrong on anything unusual. That idea runs through the whole case study.

Training Data

Three classes, six samples each

I created three classes and uploaded six pose samples to each, 18 images in total. After adding the samples I pressed Train Model, and the status changed to 'Model Trained'.



The three classes with 6 samples each in Teachable Machine.

6 samples
One or both hands raised

Hand Raised

6 samples
Person seated on a chair or stool

Sitting

6 samples
Person standing upright

Standing

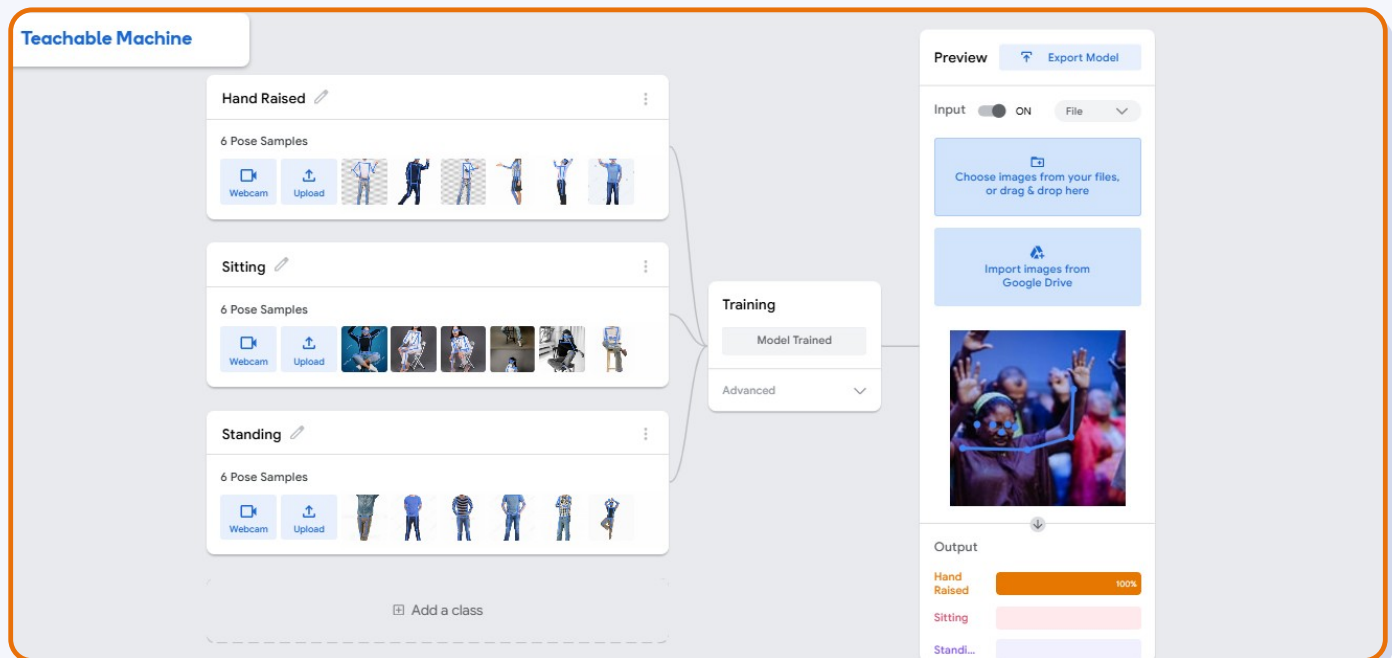
What I notice in the training images

Several samples are cut-outs with plain or transparent backgrounds, and the pose skeleton is drawn on many of them. That makes learning easier, but real photos have busy backgrounds, so the model may behave differently outside this setup.

Testing the Model

New images, never seen in training

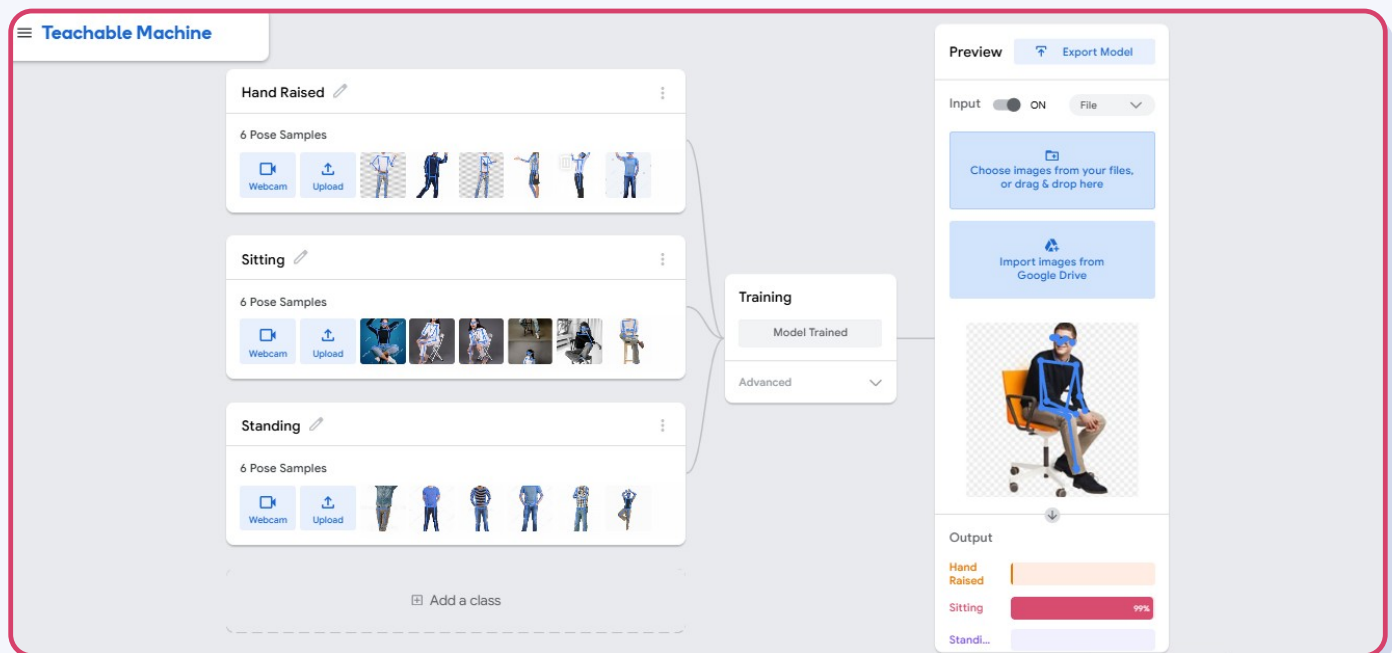
Teachable Machine's Preview panel shows the model's prediction as coloured bars: orange for Hand Raised, pink for Sitting, purple for Standing. The Output bars at the bottom right of each screenshot show the result.



Test 1: a new image of a person with raised hands

Predicted Hand Raised, 100%

Correct



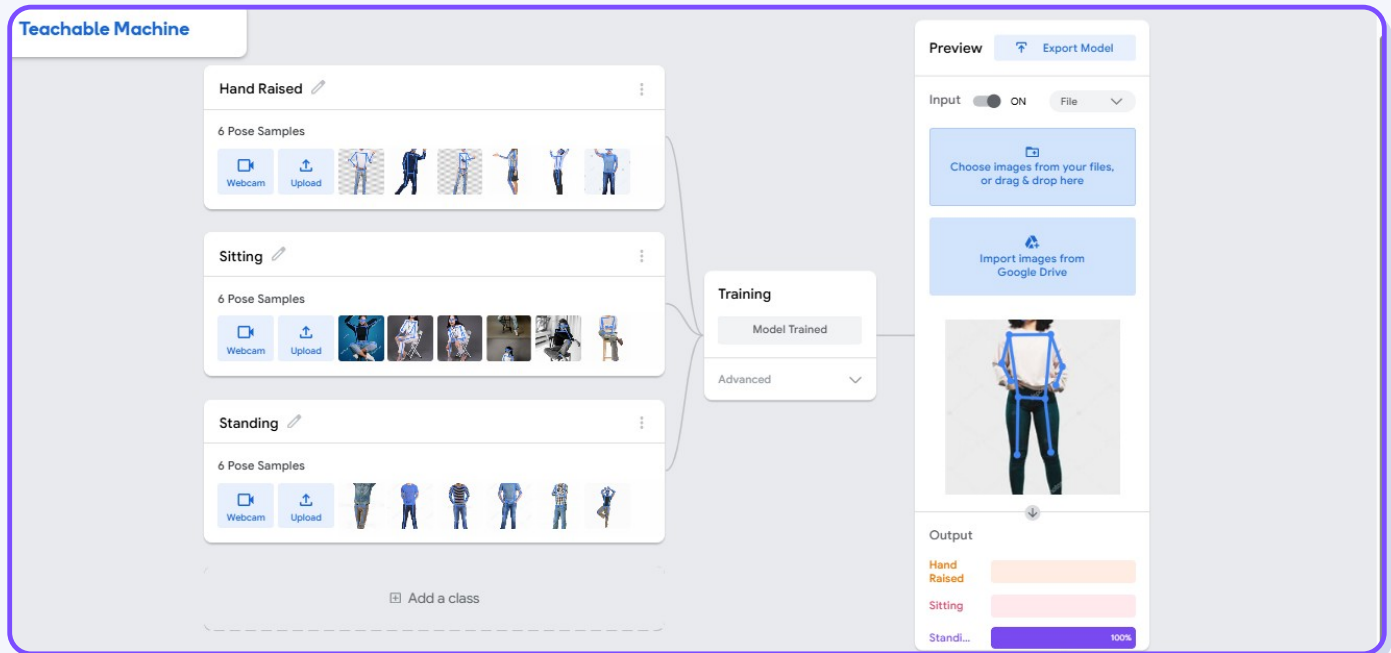
Test 2: a new image of a man sitting on a chair

Predicted Sitting, 99%

Correct

Test 3 and Results Summary

Three unseen images, three correct predictions



Test 3: a new image of a woman standing

Predicted Standing, 100%

Correct

Results at a glance

Test image	Model predicted	Confidence	Result
Hand Raised	Hand Raised	100%	Correct
Sitting	Sitting	99%	Correct
Standing	Standing	100%	Correct

3 of 3 new images classified correctly

A good start. The next page explains why this is not yet proof of a reliable model.

Analysis: What the Results Mean

A product manager's reading of a small test

What worked

- + The model learned from only 18 images and no code.
- + All three unseen test images were predicted correctly.
- + Confidence was high on every test (99% to 100%).
- + Teachable Machine makes it cheap to retrain and try again.

What this test cannot tell us

Only one test image per class

Three images is a very small sample. A lucky streak and a reliable model look the same at this size.

Med

High confidence is not accuracy

100% means the model is sure, not that it is right. Models can be confidently wrong on unusual images.

Med

The classes overlap

'Hand Raised' is an action while 'Sitting' and 'Standing' are postures. A person sitting with a raised hand fits two classes. Separate models or clearer classes would help.

Med

Clean training images

Several samples are cut-outs on plain backgrounds, so cluttered real-world photos are untested.

Low

Bottom line

A promising first model, not yet evidence of a reliable one. The next step is a larger, more varied test set.

Product Thinking

Where a model like this could be useful

A pose classifier is a building block, not a product. These are ideas I would explore, and each one needs user research before anyone builds it.

Online tutoring

Notice when a student raises a hand on video, so a tutor does not miss them in a busy class.

Posture reminders

Nudge desk workers when they have been sitting for a long time.

Exercise form

Check basic poses in a home workout or physiotherapy routine.

Classroom tools

Help teachers see class participation patterns, with consent.

Questions an AI product manager should ask

Who is the user, and what do they need?

A tutor wants fewer missed hands, not a technology demo.

What happens when the model is wrong?

A missed hand raise is a small annoyance. A wrong medical or safety call is not.

Whose images are being used?

Camera feeds of students, especially minors, need clear consent and privacy rules.

How good is good enough?

Set a target (for example, accuracy on 10 or more new images per class) before building more.

Next Steps and Lessons

How I would make the model more reliable

1

Add more varied samples

Different angles, lighting, backgrounds, clothing and people.

2

Add an 'Other / no person' class

So the model can say 'none of these' instead of guessing.

3

Build a proper test set

10 or more new images per class, and record every result in a table.

4

Try a live webcam test

See how the model behaves in real time with movement.

5

Clarify the classes

Separate the 'hands' question from the 'sitting or standing' question.

Skills this project shows

Data labelling

Model testing

Reading results critically

Product thinking

Responsible AI questions

What I learned

A working model can be built without code, but a working demo is not the same as a reliable product. Good results on three images are a promising start. Real confidence comes from more varied data, a larger test set and clear classes.

Salma Mazhar

Message me to discuss an AI or EdTech project.